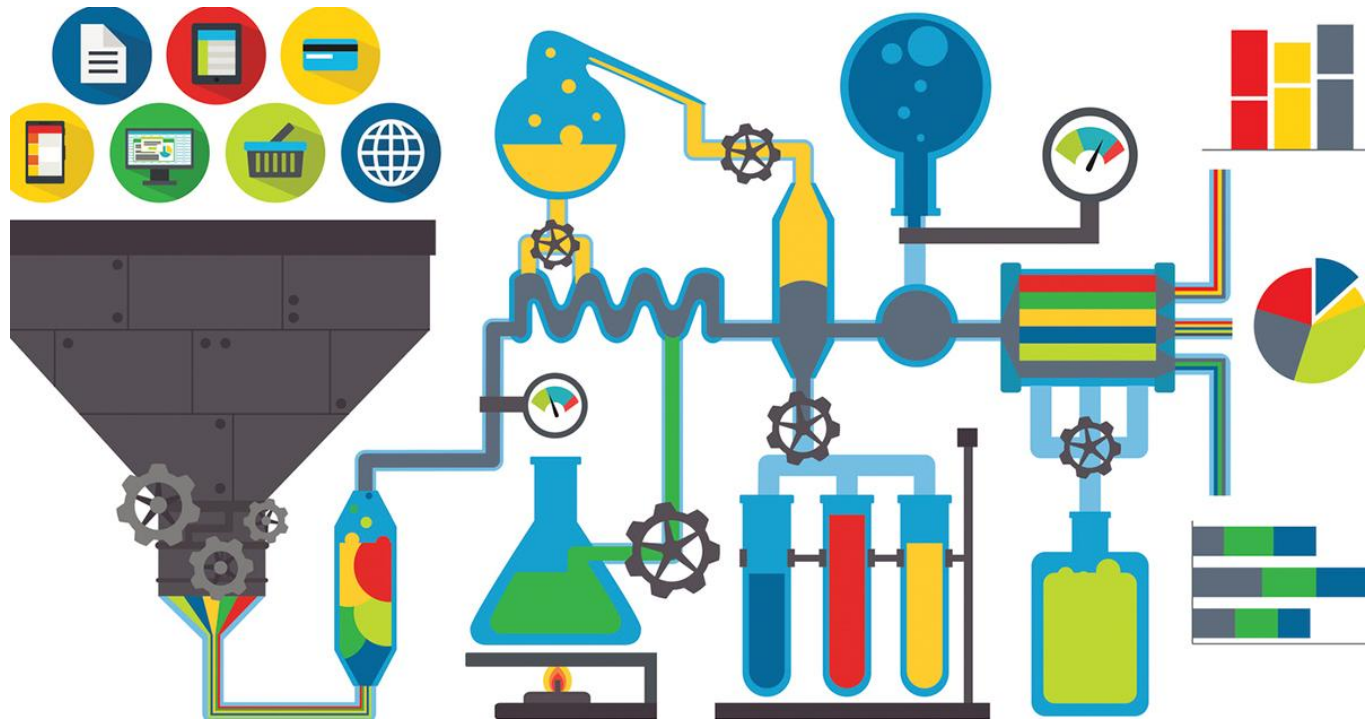




LECTURE – I

DATA WAREHOUSES

SCOPE





ORGANISATIONAL NEEDS FOR DATA HAVE EVOLVED AROUND TWO MAIN PURPOSES:

DATA CAPTURE

DATA ANALYSIS

HOW ORGANISATIONAL NEEDS FOR DATA HAS EVOLVED AROUND TWO MAIN PURPOSES.

- Data Capture
 - it all started with the sole objective of capturing transactional data for operational records
 - customers' contact details, sales orders, invoices, payments, product details and so on.
 - database were created to serve 2 purposes:
 - Record and keep track of business activities quickly and digitally
 - (so that we don't have to deal with papers flying everywhere and getting lost forever)
 - Help business users to complete their day-to-day tasks more effectively
 - (e.g. whether an order has been shipped or how much inventory we are having)
 - Simply put, focusing on the present and making sure business transactions happen as smoothly as possible



DATA, DATA IS EVERYWHERE

- First database solutions data back to the '60
- Enterprises are adept at gathering all sorts of data
 - about their customers, prospects, internal business processes, suppliers, partners, and even competitors

SHORT HISTORY

- What Goes Around Comes Around, 2006
 - Joseph Hellerstein (a computer science professor at UC Berkeley)
 - Michael Stonebraker (founder of Ingres and Postgres)
- The paper takes us through eras of data models:
 - Hierarchical (IMS): Late 1960s and 1970s
 - Network (CODASYL): 1970s
 - Relational: 1970s and early 1980s
 - Entity-Relationship: 1970s
 - Extended relational: 1980s
 - Semantic: Late 1970s and 1980s
 - Object-oriented: Late 1980s and early 1990s
 - Object-relational: Late 1980s and early 1990s
 - Boring days – 1990s ☺
 - some rewrites mSQL→MySQL, Sybase→SQL Server, Ingres→Postgres(QL)
 - new idea – analytical needs
 - Semi-structured (XML): Late 1990s to the present

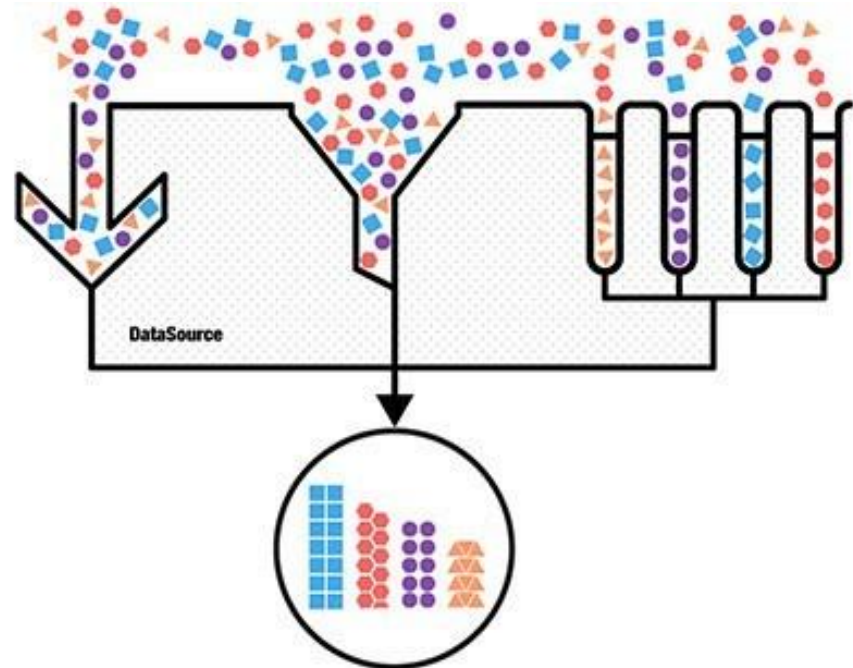


DATA

- Capturing data, however, is just the beginning
 - BARC's BI Trend Monitor 2020 studied perceived importance of Data-Driven Culture in companies (2637)
 - Data-driven culture was identified as the third most important trend, rising from fifth place in the previous year.
 - Creating a data-driven culture is about replacing gut feeling with decisions based on data-derived facts
 - simple key figures such as revenue or profit, results from advanced analytics models, or even qualitative data
 - There are some preconditions in becoming data-driven

DATA

- Accounting software, resource planning, logistics, production, project or CRM systems - these are just some of the IT solutions that are used in companies on a daily basis.
- Each such system collects huge amounts of data, which together can be a valuable source of business knowledge – provided that they are skillfully aggregated, processed and presented



SHORT HISTORY

- The paper takes us through nine eras of data models:
 - 2000s – INTERNET BOOM
 - DATA WAREHOUSES
 - MAPREDUCE SYSTEMS
 - NoSQL SYSTEMS
 - 2010s
 - NewSQL SYSTEMS
 - CLOUD SYSTEMS
 - SHARED-DISK ENGINES
 - GRAPH SYSTEMS
 - TIMESERIES SYSTEMS
 - 2020s – Web 3.0 and AI times
 - BLOCKCHAIN DATABASES
 - VECTOR DATABASES



ANALYSIS IS WHAT WE NEED

- Late '90s

- This is the time we will focus on, as then some traditional DW solutions were introduced



HOW ORGANISATIONAL NEEDS FOR DATA HAS EVOLVED AROUND TWO MAIN PURPOSES.

- Data Analysis
 - Businesses quickly realised living in the moment wasn't good enough.
 - New questions arose.
 - How are we performing as compared to last month?
 - Why was there a fall in revenue in the last two quarters?
 - What have our customers regularly complained about since the beginning of the year?
 - The need to capture all operational records is still indispensable, but there is now another equally important objective: analysing historical data for decision-making.
 - Unfortunately, a database once built to capture transactional data wasn't suitable to empower management with enough facts and historical insights for making informed decisions.
 - vast difference between capturing present records and storing historical data across a variety of data sources for analysis, we need another tool.

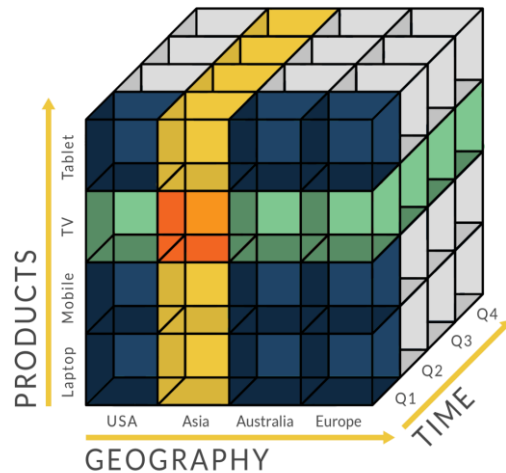
DATA VS NEEDS

- Let's imagine a situation in which a company has three systems:
 - Finance & Accounting
 - About customer invoices
 - Manufacturing
 - Information about production costs related to the implementation of projects for our customers
 - CRM
 - Information about the costs of commercial activities (customer acquisition).
- How to create a report comparing the revenues from a given client with the expenses that the company must incur in connection with its service?



BUSINESS ADVANTAGE

- You have a need for specific data, but the system it's on won't let you access it in the way you need. This is why data warehouses are created.
 - N.Mannon, Blastam
- Information is the fundamental building block of decision process
 - It is important to deliver adequate information to decision makers



WE NEED SPECIFIC INFORMATION

Design led data & KPI visualization

- Metrics that matter
- Operations management dashboards
- Supervisory level dashboards



Machine learning based approach

- Exploratory data analytics
- Vision analytics – inspection automation, surveillance etc.
- Predictive analytics



Model driven business improvement approach

- Value driver tree based analysis
- What-if analysis of levers



Data Platform

- One data warehouse
- Cloud-based data repository
- IoT based data platforms

Real time data analytics

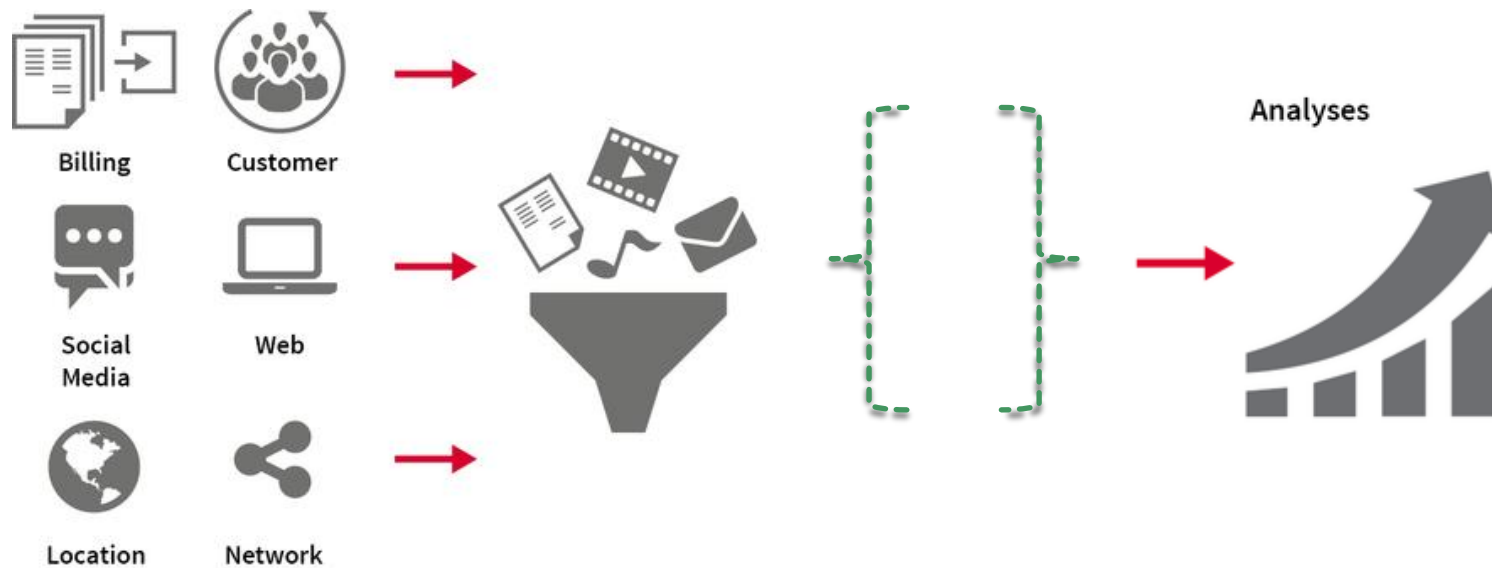
- Sensor data acquisition using IoT based platforms
- Streaming analytics & complex event processing



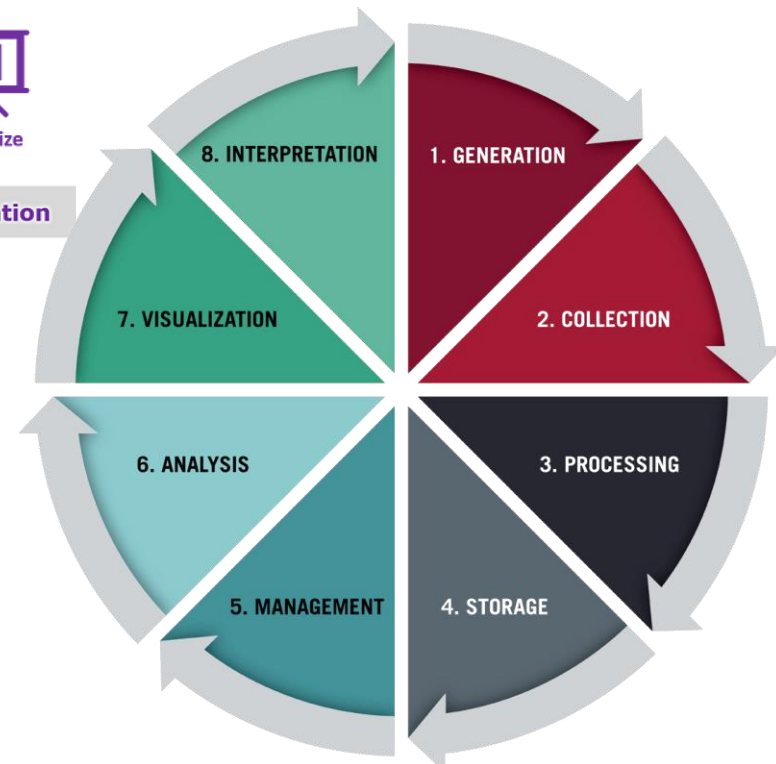
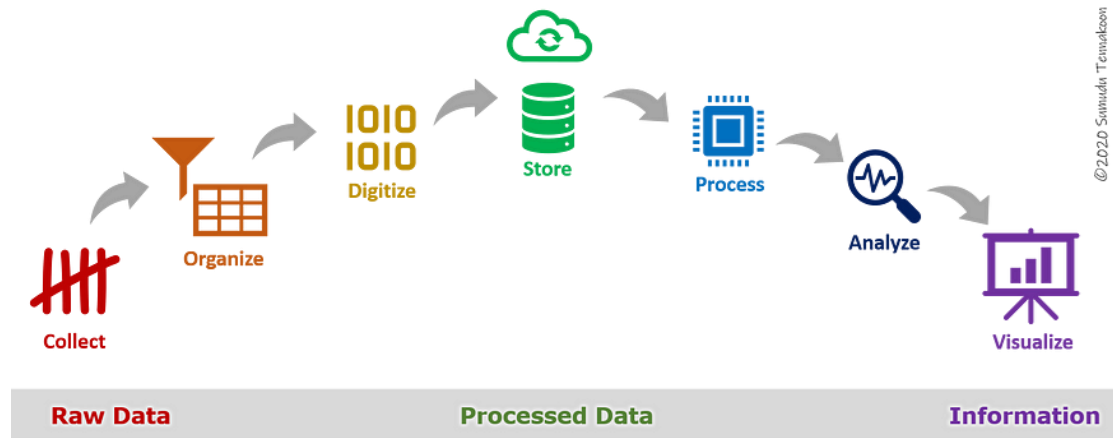
USER NEEDS VS. SOURCE DATA

- Let's imagine a situation in which a company has three systems:
 - Finance & Accounting
 - About customer invoices
 - Revenue from the client
 - Manufacturing
 - Information about production costs
 - CRM
 - Information about the costs of commercial activities
 - (like customer acquisition, customer support)
- Creating a basic report like:
 - Comparing the revenues from a given client with the expenses that the company must incur in connection with its service:
- Typically, this task is troublesome:
 - You need to export filtered data from all three systems, integrate and clean this data, and finally prepare a report for the manager
- Typically, this repeats
 - This procedure must be repeated when the data in one of the source systems is updated

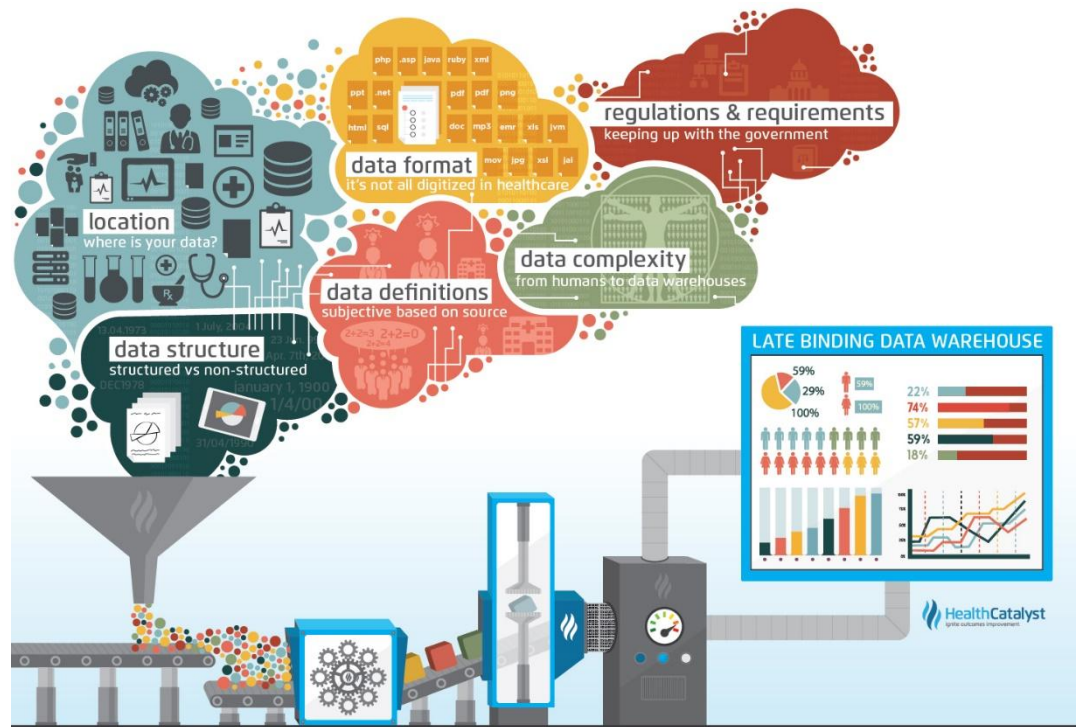
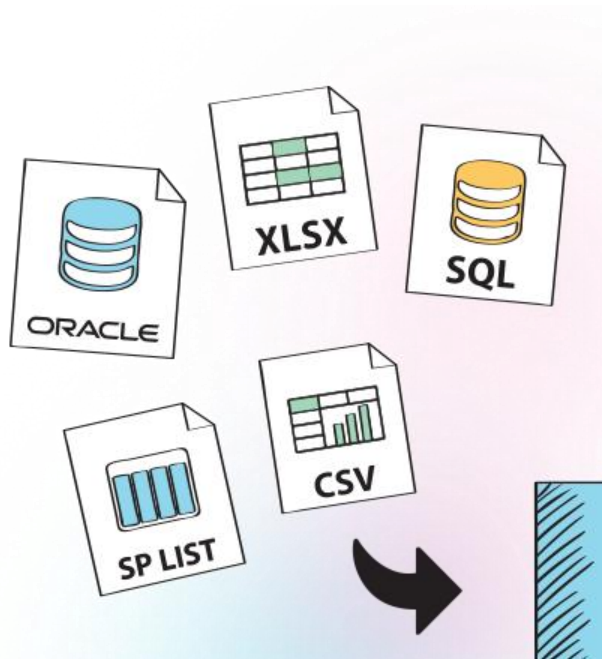
SCOPE



FROM DATA TO INFORMATION



DATA



DATA IS NOT ENOUGH

- Heterogeneous applications, unsystematic format and poor data quality make it difficult to analyse the data
 - impede decision making

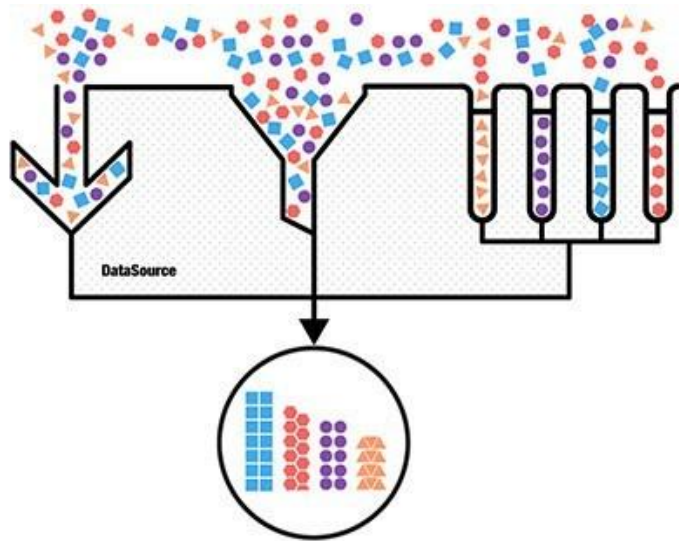




DATA ISSUES

- Bad customer data costs companies six percent of their total sales
 - according to a UK Royal Mail survey
 - <https://www.royalmail.com/business/system/files/royal-mail-data-services-insight-report-2018.pdf>
- The overall cost of poor data quality for U.S. businesses at \$3.1 trillion per year
 - estimate by IBM
 - <https://www.entrepreneur.com/article/393161>
- The UK Government's Data Quality Hub estimates organizations spend between 10% and 30% of their revenue tackling data quality.

WE NEED INFORMATION

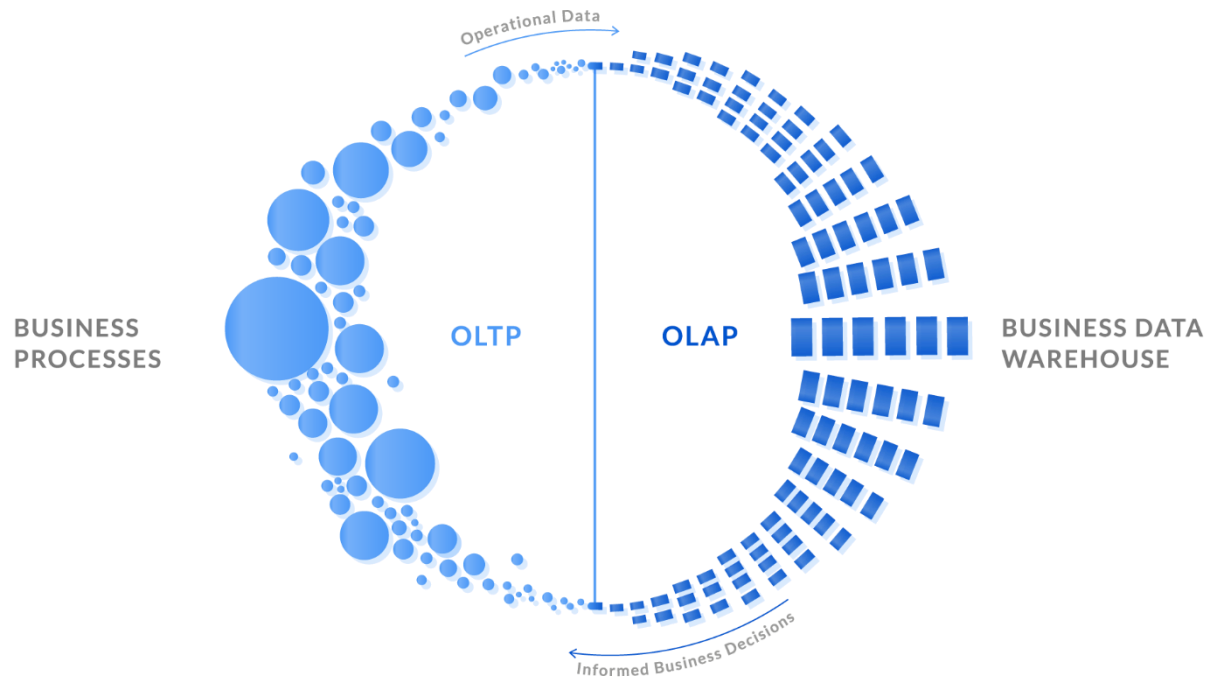


- Information
 - Fundamental element of the decision support process
 - Allows for competitive edge
 - Human intuition and hunches are not enough for optimal decisions
- Information delivery is possible, when
 - Proper efficiency/quality measures are defined
 - Proper questions are formulated
 - Did something happened?
 - Why something happened?
 - What is going to happen?

EXAMPLE

- Consider three systems in an org.
 - Accounting and finances
 - invoices
 - Production
 - production costs
 - CRM
 - Marketing costs
- Creating a report that puts together inflows and outflows for a given customer
 - Is not trivial task
 - Filter data, export – all three systems
 - Integrate and clean data
 - Create a report
 - Data refresh requires a redo

DIFFERENT NEEDS

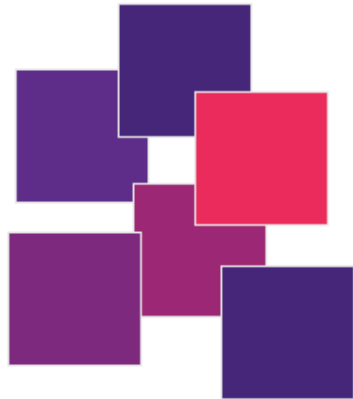


DATA WAREHOUSE

- “Through measurement comes knowledge.”
 - Heike Kamerlingh Onnes
- A data warehouse is an example of the journey that data takes, when combined with context, to become information.
 - Prior to application of context it is just a collection of numbers and letters, bits and bytes.
 - Yet information is still not enough to enable an organization to learn from and act based on what they have collected.
 - The ability to use accurate data and timely information to objectively measure and, therefore, proactively manage outcomes and business processes demonstrates the value of a data warehouse.

DATA WAREHOUSE

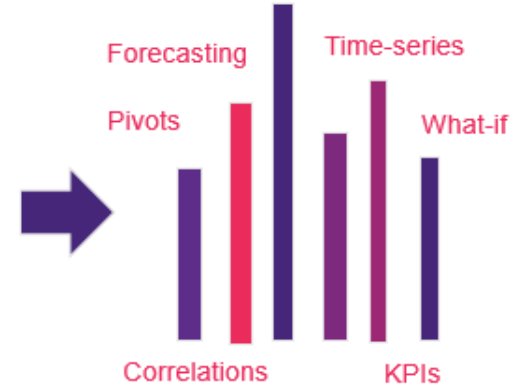
Unorganized data in ERP,
CRM, and other systems



Pulled into a structured
data warehouse



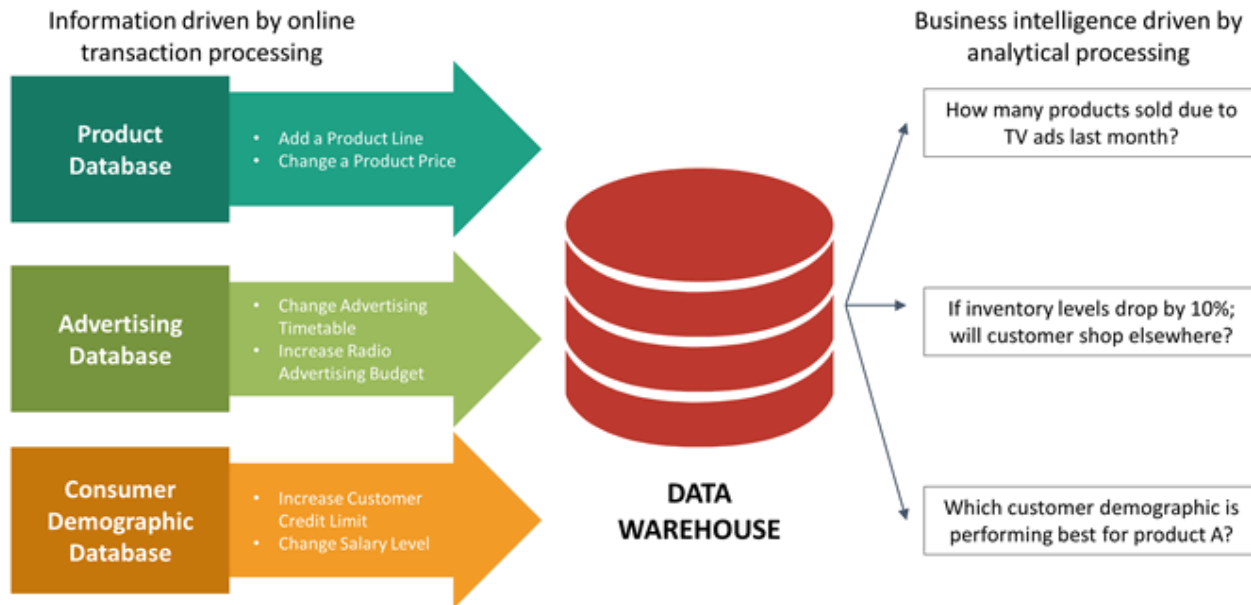
Analyzed in a BI
application



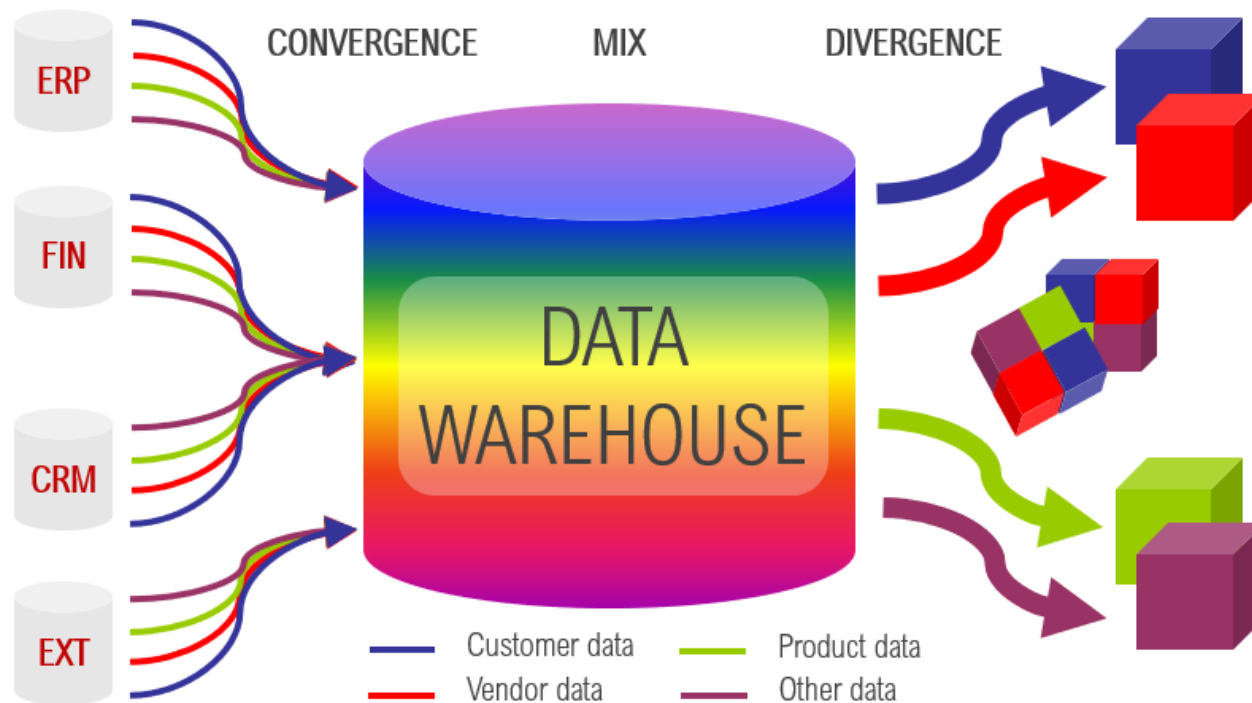
DATA WAREHOUSE

BUSINESS INTELLIGENCE

Business Intelligence Example



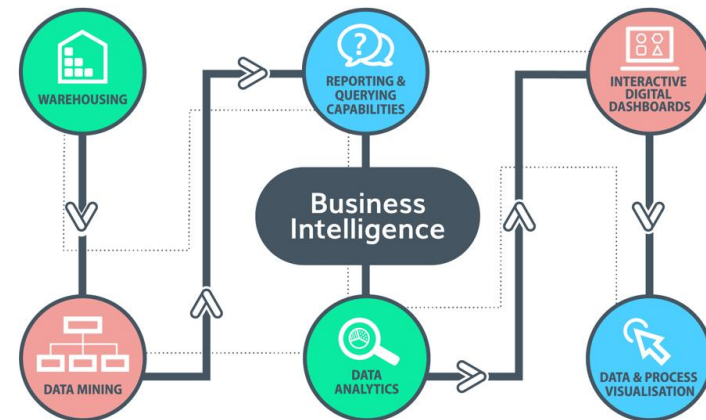
DATA WAREHOUSE



BUSINESS INTELLIGENCE

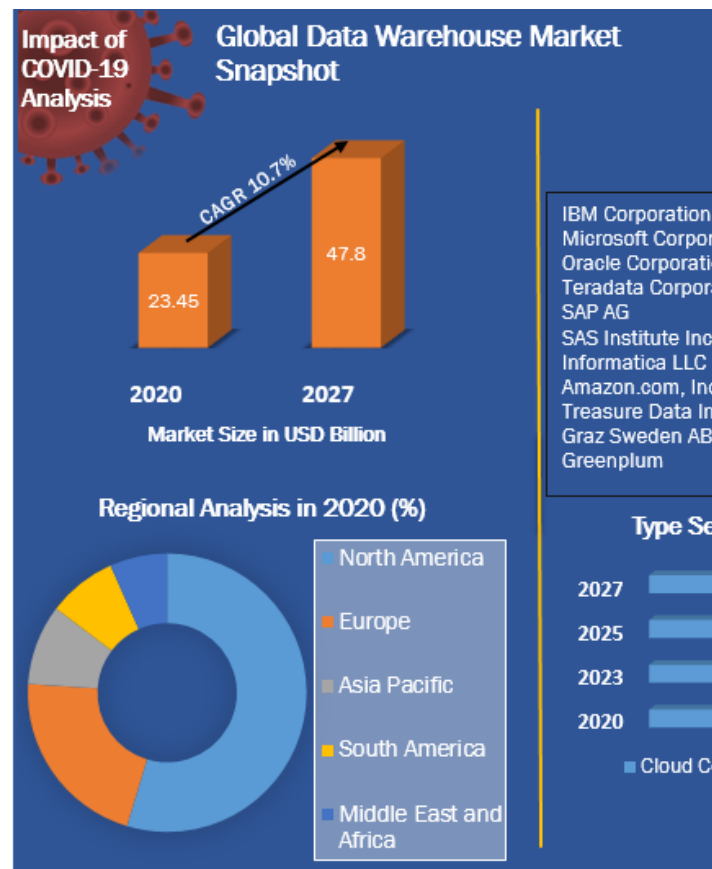
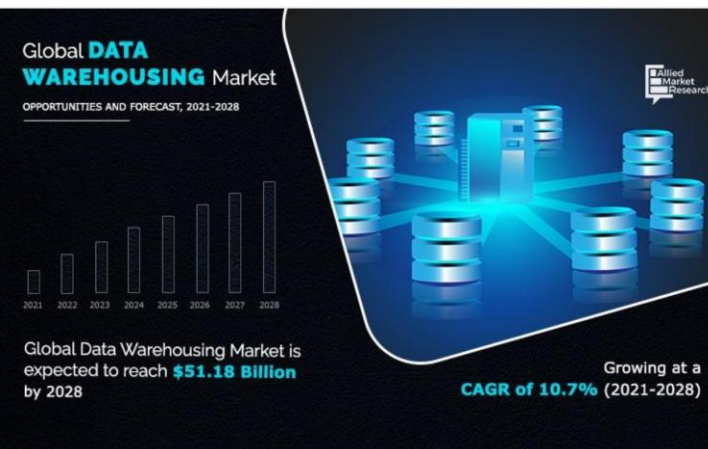
source: [saksoft.com](https://www.saksoft.com)

- Business Intelligence systems
 - are tools, technologies and applications which support data integration, analysis and reporting.
 - turning raw data into actionable insights for better business decisions.
 - methodologies, processes, architectures, and technologies that transform raw data into meaningful and useful information for business purposes.
- Data warehouses
 - have become a cornerstone of modern data management, especially in the realm of Business Intelligence (BI) and analytics.
 - popularity stems from their ability to solve a critical problem:
 - how to effectively use vast amounts of data for informed decision-making.



DATA MANAGEMENT AND SYSTEM SECURITY

- When building a BI system, you are not only managing bits, but also trust in data.
- Key concepts:
 - Data lineage
 - Data traceability (crucial for audits and errors in reports)
 - Data quality
 - Automatic data quality testing in pipelines.
 - Metadata Management
 - Data dictionary – what does a given column in the system mean?
 - Security
 - GDPR, sensitive data masking, row-level security.



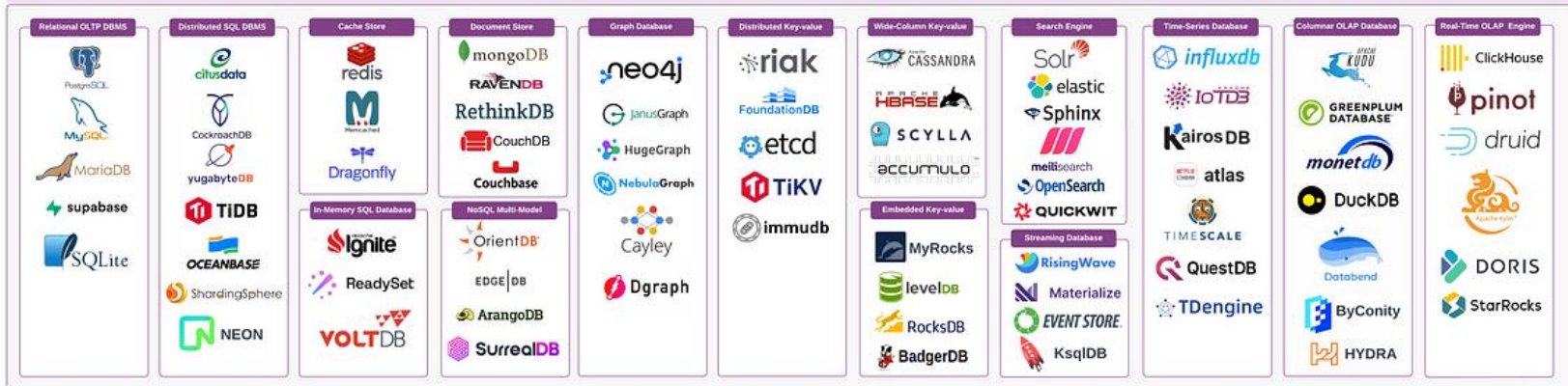
MARKET

- Global Data Warehouse Market size was valued at US\$ 23.45 Bn. in 2020
- According to the latest report by IMARC Group the market is expecting to grow at a CAGR of around 11% during 2021-2026.
 - "Data Warehousing Market: Global Industry Trends, Share, Size, Growth, Opportunity and Forecast 2021-2026,,
- Similarly by Allied Market Research:
 - „Data Warehousing Market by Type of Offering (ETL Solutions, Statistical Analysis, Data Mining, and Others), Type of Data (Unstructured and Semi-Structured & Structured), Deployment Model (On-Premise, Cloud, and Hybrid), Enterprise Size (Large Enterprises and Small & Medium Enterprises), and Industry Vertical (BFSI, IT & telecom, Government, Manufacturing, Retail, Healthcare, Media & Entertainment, and Others): Global Opportunity Analysis and Industry Forecast, 2021-2028”
- New BI reports suggest:
 - „The global business intelligence market size was valued at USD 29.42 billion in 2023 and is projected to grow from USD 31.98 billion in 2024 to USD 63.76 billion by 2032, exhibiting a CAGR of 9.0% during the forecast.”
<https://www.fortunebusinessinsights.com/business-intelligence-bi-market-103742>

OPEN SOURCE DATA ENGINEERING LANDSCAPE 2024



STORAGE SYSTEMS



DATA LAKE PLATFORM



DATA INTEGRATION



DATA PROCESSING & COMPUTATION



WORKFLOW & DATAOPS



DATA INFRASTRUCTURE & MONITORING



ML/AI PLATFORM



METADATA MANAGEMENT



ANALYTICS & VISUALISATION

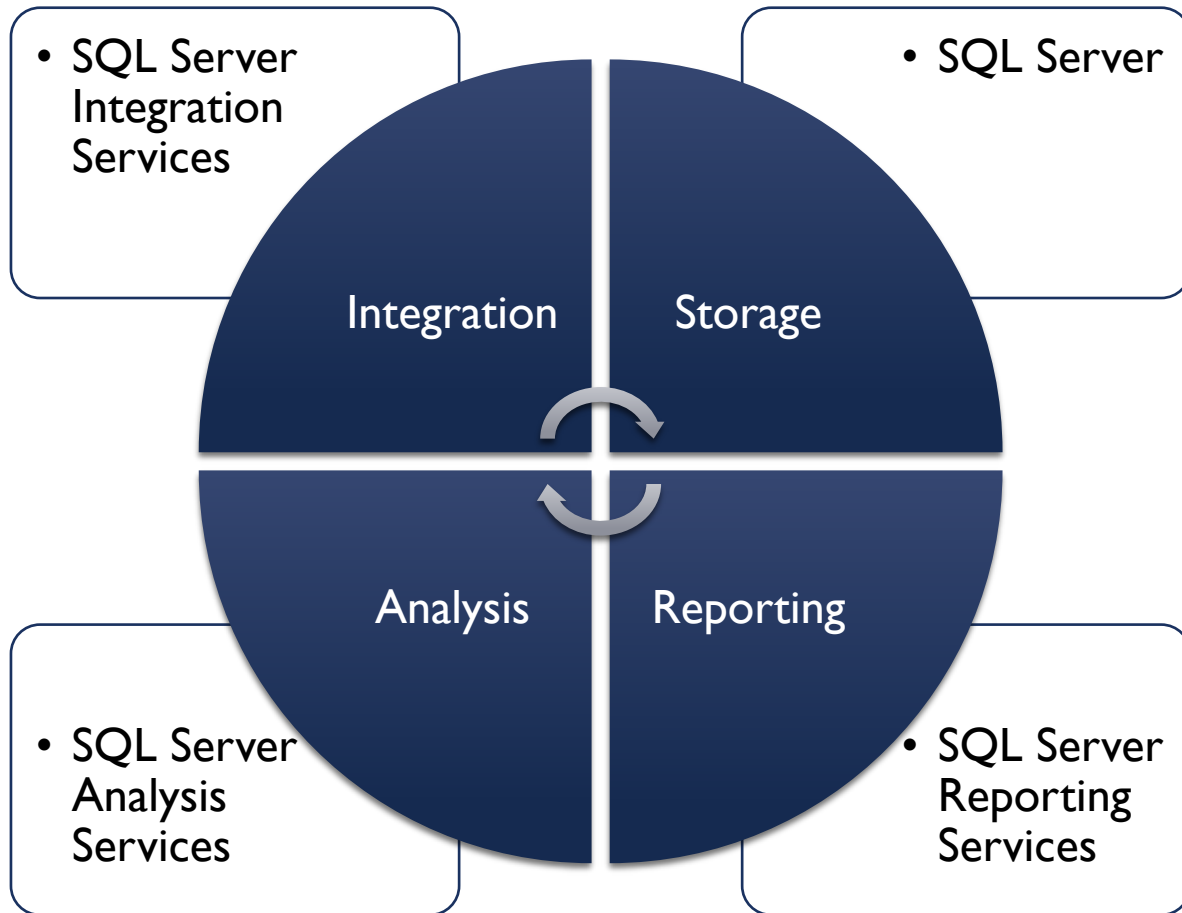


Figure 1: Magic Quadrant for Analytics and Business Intelligence Platforms



BI

SQL SERVER ECOSYSTEM



By avoiding left turns
whenever possible,
UPS estimates to save:

10 million

gallons of fuel a year



6 to 8 fewer miles
driven per route

100,000

metric tons of CO₂
emissions a year

(equivalent to **21,000** cars
taken off the road)



UPS

AMAZON

- Amazon, being the largest online retailer, uses big data analytics in a number of ways:
 - Product Recommendation:
 - Even if you only "take a look" at the product - Amazon saves and uses this data.
 - Similar products appear on the website, in the social media channel, and in the ads.
 - Amazon earns 35% of its annual sales through product recommendations
 - Dynamic Pricing:
 - Have you ever noticed that if you check the product repeatedly, then the prices seem to go up.
 - This is because Amazon has tracked your intentions and knows that you may be willing to pay even more for the product.
 - The same logic is also used when selling airline tickets.

SPORT

- FI

- For example, the Mercedes AMG FI W08 EQ Power+ cars were equipped with 200 sensors that transmitted millions of data points during the race weekend.
- Reportedly, more than 300 GB of data is transferred from the car.
- Williams says it has more than a thousand feeds of data collected at the same time as the race.
- Red Bull's RB12 was equipped with around 100 sensors collecting data on 10,000 different components.

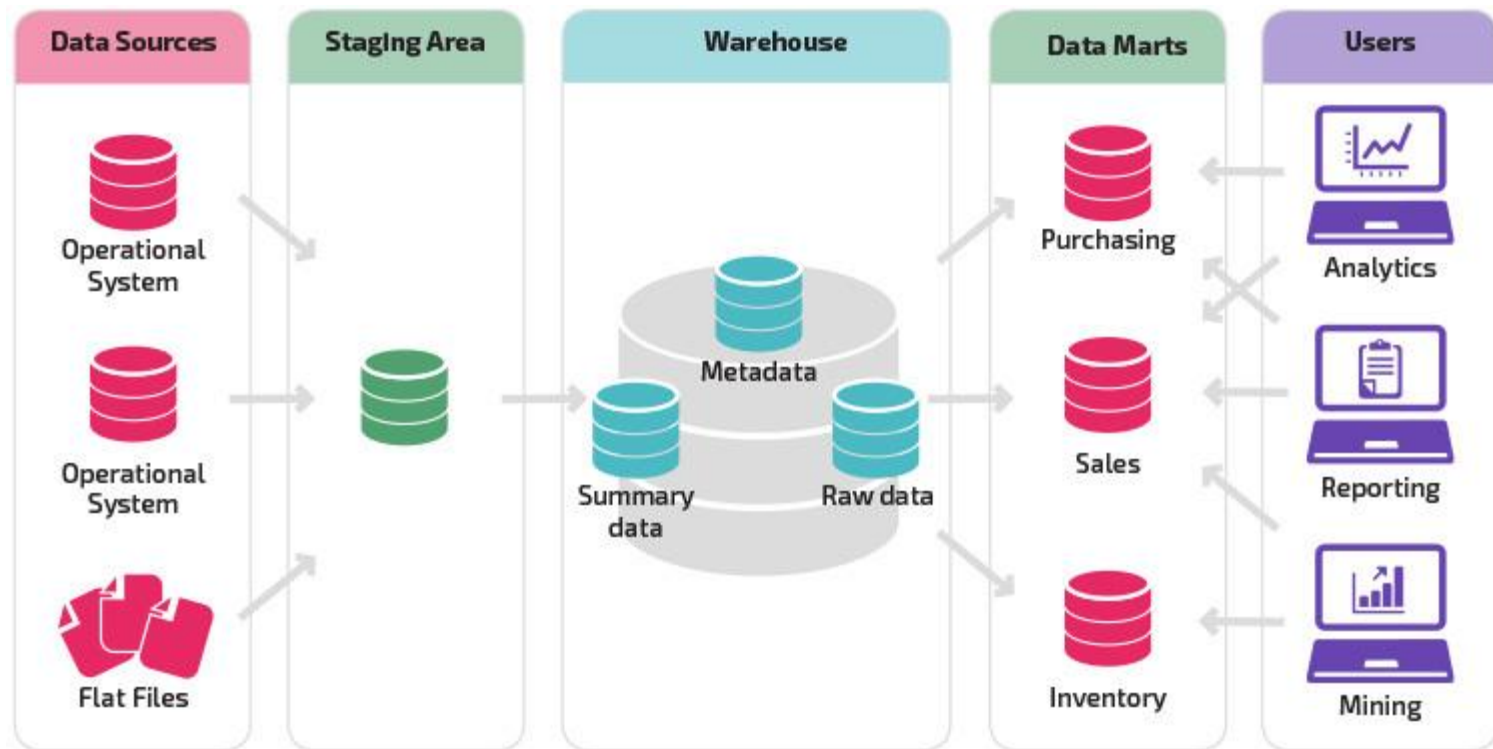
- NFL

- Using artificial intelligence and data analytics to determine how best to engage fans.
- With the increasing use of social media tools, NFL messengers are looking to attract their audience, build a fan base, and promote the public profiles of top players. Using a data-driven approach such as next-gen stats, they can track player movements, determine player speeds, and document the number of yards a player has run.
- Analytics departments were created to not only understand the data being tracked, but also to decipher data sets from other sources, such as trainers
- "There's so much data out there right now; Their job is to take everything and put it into context. "

DESIGN WORKSHOP – FI DECISION SYSTEM

- Task:
 - System for monitoring and supporting strategic decisions during a race.
- For example:
 - Imagine that you are designing a dashboard for a racing engineer.
 - You cannot show him everything – 300+ sensors.
 - You need to present certain summaries and key performance indicators (KPIs)
 - 3 key technical parameters (e.g. tyre temperature, fuel consumption, aerodynamics)
 - You need to design alarm thresholds.
 - Alarm rule (e.g. if $X > Y$ for Z seconds, send an alert to the race engineer).

DATA WAREHOUSING



MODERN TRENDS

- Modernization

- may involve previously untapped features or platforms,
 - such as in-memory functions, in-database analytics, clouds, and data types or analytics that are new to your organization.
- Besides the core warehouse, the systems and tools integrated with it need modernization too,
 - especially those for analytics, reporting, and data integration.
- Russom P., Modernizing Data Warehouse Infrastructure, TDWI Checklist Report, 2018

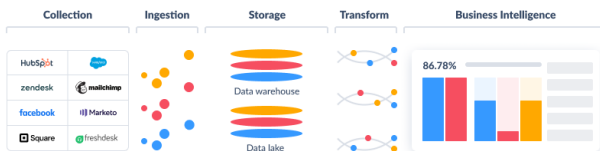


MODERN



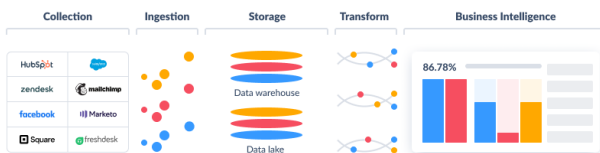
Modern Data Stack architecture

Basic Version



Modern Data Stack architecture

Advanced Version

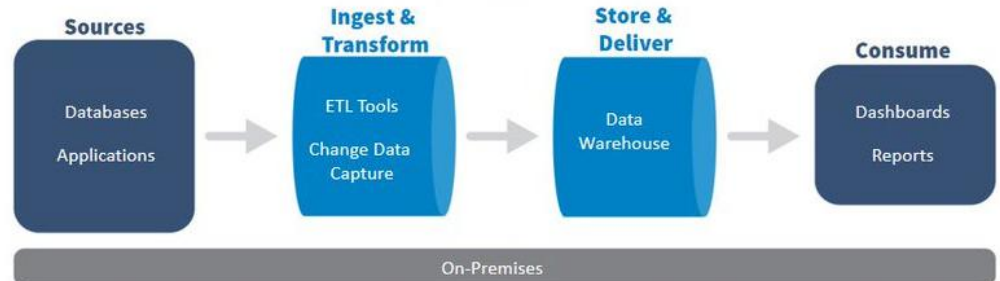


Data quality

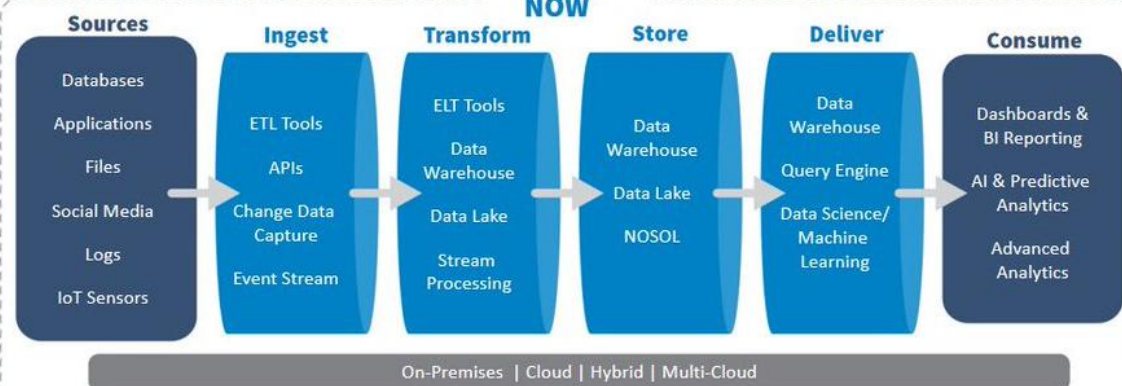
Data governance

Enterprise Data Pipelines Have Changed

THEN



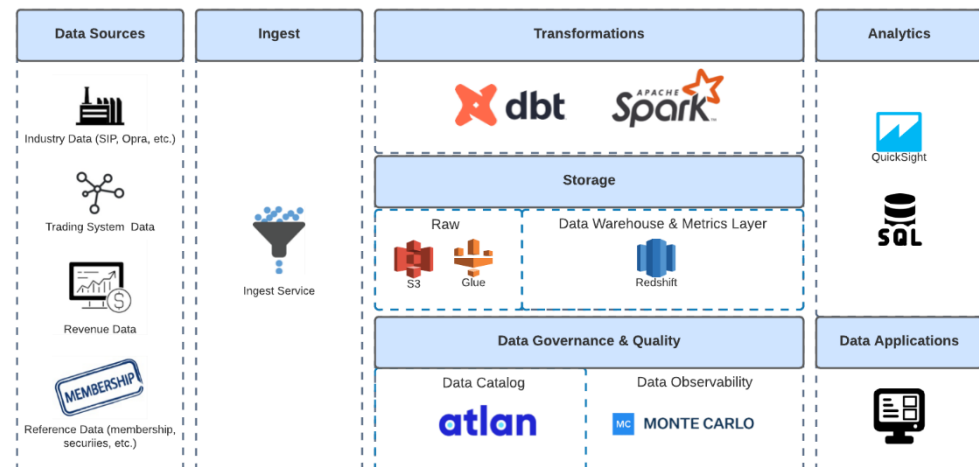
NOW



Multiplying users & use cases • Higher data volumes from many sources • More hybrid & multi-cloud platforms

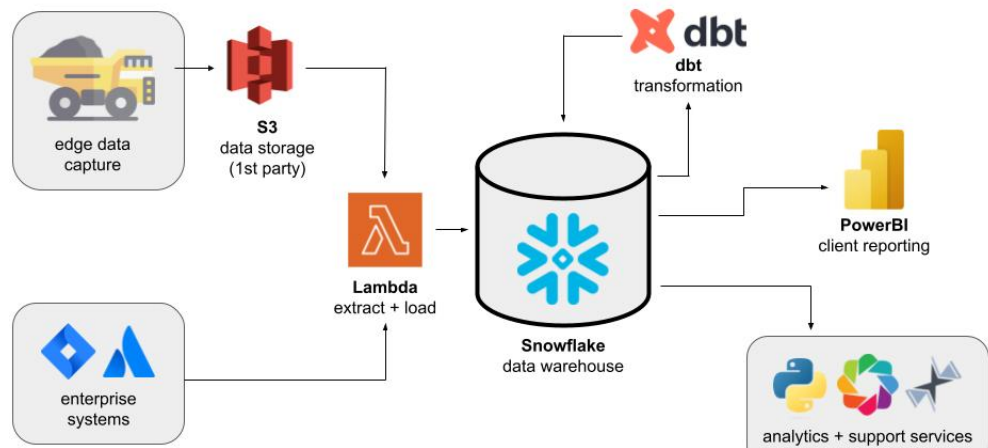
NASDAQ

- Nasdaq manages 30 stock exchanges in North America and the Nordics and is home to more than 4,000 public companies. They also provide technology to 2,200 financial institutions in 130 markets.

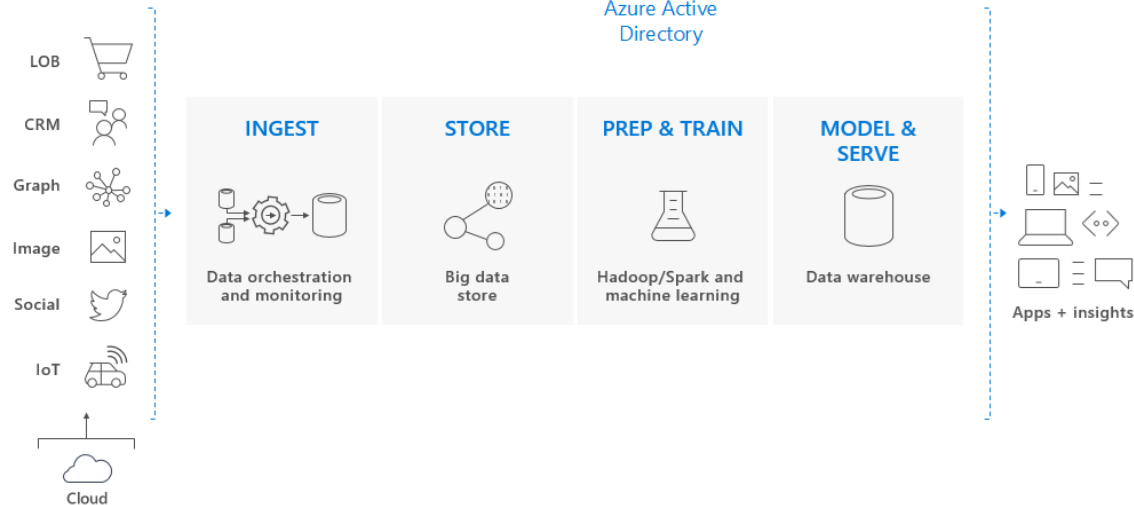
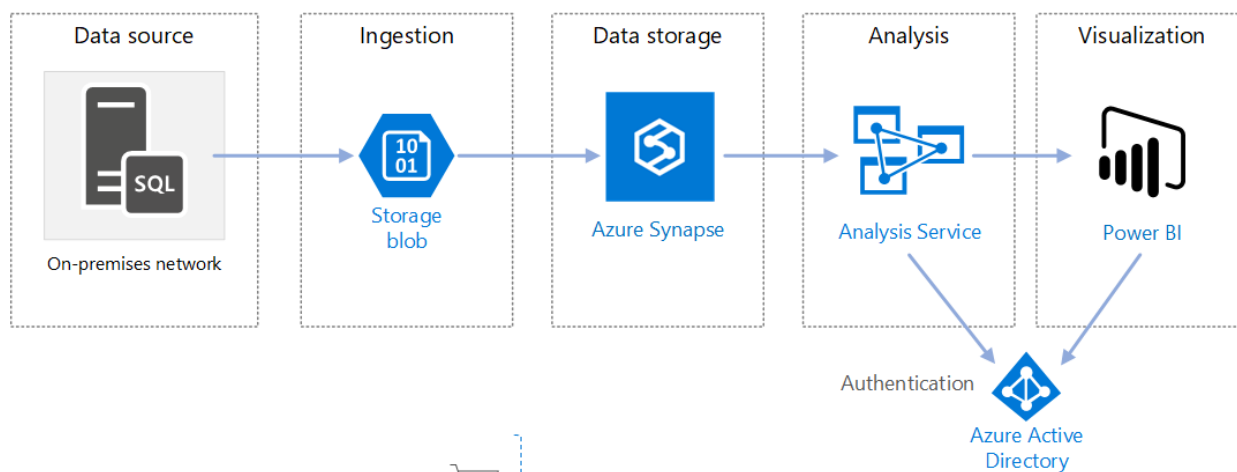


MAXMINE

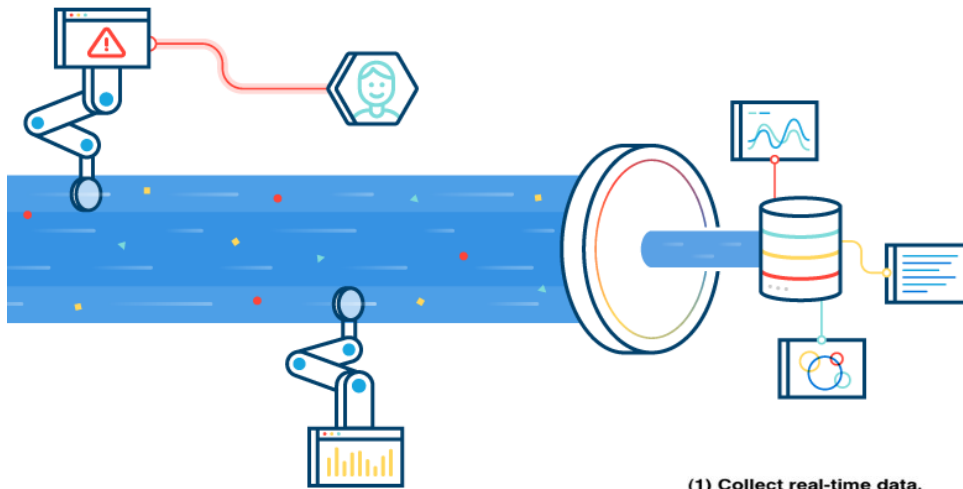
- MaxMine captures, processes and analyzes data from mining equipment in open-pit mines in Australia and Africa



MODERN TRENDS: CLOUD



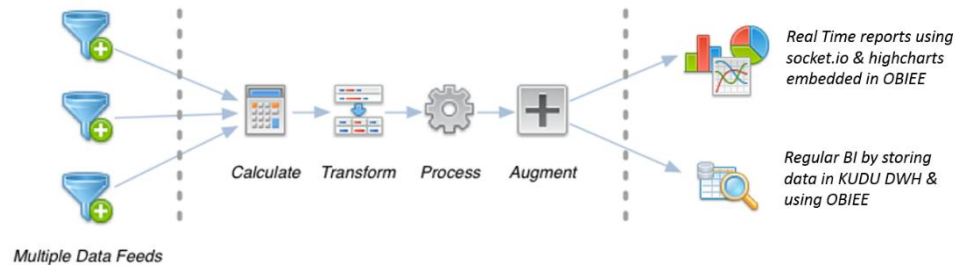
MODERN TRENDS: DW - STREAMS



(1) Collect real-time data.

(2) Process data as it flows.

(3) Explore and visualize.



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